Program Listing – Washing-Machine Control Program

WM-8 five-stage pipelined washing-machine microcontroller – ELL7282/ELL782 Introduction to Computer Architecture – Assignment 2

Source: `program.asm`. ROM image: `program.hex` (Logisim `v2.0 raw` format, one 16-bit word per address, load with *right-click ROM → Load Image*). The same words are already stored in the ROM of `washing_pipeline.circ`. The encoding is mechanical: **R** = op≪12 | ra≪9 | rb≪6, **I** = op≪12 | ra≪9 | imm8, **N** = op≪12 (`tools/asm.py` only automates this table look-up). The program has 180 words (`0x00`–`0xB3`). Unused ROM words are `0x0000` (`NOP`).

Symbols

I/O		RAM variables		Actuator patterns (`ACT`)	
`IN_STATUS`	`0xF0`	`V_TARGET`	`0x00`	`A_OFF`	`0x00` all off, unlocked
`WATER_LEVEL`	`0xF1`	`V_WASH`	`0x01`	`A_LOCK`	`0x20` DOOR_LOCK
`TEMP`	`0xF2`	`V_SPIN`	`0x02`	`A_FILL`	`0x21` LOCK + VALVE_IN
`DRUM_RPM`	`0xF3`	`V_DRAIN`	`0x03`	`A_WASH`	`0x24` LOCK + MOTOR_EN (+`0x10` DIR)
`ACT`	`0xF4`	`V_TAG`	`0x04`	`A_DRAIN`	`0x28` LOCK + PUMP_OUT
`DISPLAY`	`0xF5`	`V_TEMP`	`0x05`	`A_SPIN`	`0x2C` LOCK + PUMP + MOTOR_EN
`T_COUNT/LIMIT/CTRL/STATUS`	`0xF8`–`0xFB`			`A_DONE`	`0x40` BUZZER, unlocked

Mode parameters (`MODE[0]` selects, `MODE[1]` ignored)

Mode	fill: `WL` ≥	wash	spin	drain: `WL` ≤	DISPLAY tag (MODE)
NORMAL (`MODE`=00 or 10)	100 (`0x64`)	64 ticks	32 ticks	10	`0x0`x (00) / `0x2`x (10)
HEAVY (`MODE`=01 or 11)	160 (`0xA0`)	96 ticks	48 ticks	10	`0x1`x (01) / `0x3`x (11)

DISPLAY low nibble: 1 idle, 2 lock/check, 3 fill, 4 wash, 5 drain, 6 spin, 7 stopping, 8 done; `0xE1` door-open abort, `0xEE` safety trip. High nibble = `MODE`≪2 (snapshot).

Address / hex / assembly table

Addr	Hex	Fmt	Label	Assembly	Comment
00	1000	I	reset	LDI R0, A_OFF	
01	1EF4	I		LDI R7, ACT	
02	71C0	R		ST R0, [R7]	all actuator requests off, door unlocked
03	1EFA	I		LDI R7, T_CTRL	
04	1202	I		LDI R1, 0x02	
05	73C0	R		ST R1, [R7]	timer: clear, disabled
06	1EF5	I	idle	LDI R7, DISPLAY	
07	1201	I		LDI R1, 0x01	
08	73C0	R		ST R1, [R7]	DISPLAY = 01 (ready)
09	1EF0	I	idle_wait	LDI R7, IN_STATUS	
0A	63C0	R	wait_start	LD R1, [R7]	one status snapshot
0B	1403	I		LDI R2, 0x03	
0C	5440	R		AND R2, R1	START & DOOR_CLOSED bits
0D	84FD	I		ADDI R2, -3	0 only if START=1 and DOOR_CLOSED=1
0E	A40A	I		BNEZ R2, wait_start	
0F	160C	I		LDI R3, 0x0C	
10	5640	R		AND R3, R1	MODE<<2
11	36C0	R		ADD R3, R3	
12	36C0	R		ADD R3, R3	MODE<<4 = display tag
13	1E04	I		LDI R7, V_TAG	
14	77C0	R		ST R3, [R7]	RAM[4] = tag (MODE snapshot)
15	1404	I		LDI R2, 0x04	
16	5440	R		AND R2, R1	MODE[0]
17	A41C	I		BNEZ R2, heavy	
18	1864	I		LDI R4, 100	NORMAL: fill target
19	1A40	I		LDI R5, 64	wash ticks
1A	1C20	I		LDI R6, 32	spin ticks
1B	B01F	I		JMP save	
1C	18A0	I	heavy	LDI R4, 160	HEAVY : fill target
1D	1A60	I		LDI R5, 96	wash ticks
1E	1C30	I		LDI R6, 48	spin ticks
1F	1E00	I	save	LDI R7, V_TARGET	
20	79C0	R		ST R4, [R7]	RAM[0] = target
21	8E01	I		ADDI R7, 1	
22	7BC0	R		ST R5, [R7]	RAM[1] = wash
23	8E01	I		ADDI R7, 1	

Addr	Hex	Fmt	Label	Assembly	Comment
24	7DC0	R		ST R6, [R7]	RAM[2] = spin
25	8E01	I		ADDI R7, 1	
26	180A	I		LDI R4, 10	
27	79C0	R		ST R4, [R7]	RAM[3] = drain threshold
28	1CF4	I		LDI R6, ACT	R6 = actuator latch pointer (kept)
29	1220	I		LDI R1, A_LOCK	
2A	7380	R		ST R1, [R6]	
2B	1EF5	I		LDI R7, DISPLAY	
2C	22C0	R		MOV R1, R3	
2D	8202	I		ADDI R1, 0x02	
2E	73C0	R		ST R1, [R7]	DISPLAY = tag|2
2F	1EF0	I		LDI R7, IN_STATUS	
30	63C0	R		LD R1, [R7]	
31	1402	I		LDI R2, 0x02	
32	5440	R		AND R2, R1	
33	94A0	I		BEQZ R2, door_abort	lock request does not prove the door is shut
34	1221	I		LDI R1, A_FILL	
35	7380	R		ST R1, [R6]	
36	1EF5	I		LDI R7, DISPLAY	
37	22C0	R		MOV R1, R3	
38	8203	I		ADDI R1, 0x03	
39	73C0	R		ST R1, [R7]	DISPLAY = tag|3
3A	1E00	I		LDI R7, V_TARGET	
3B	69C0	R		LD R4, [R7]	R4 = target
3C	1EF1	I		LDI R7, WATER_LEVEL	
3D	63C0	R	fill	LD R1, [R7]	
3E	C300	R		SGEU R1, R4	R1 = (WATER_LEVEL >= target)
3F	923D	I		BEQZ R1, fill	
40	1220	I		LDI R1, A_LOCK	
41	7380	R		ST R1, [R6]	close valve
42	1EF5	I		LDI R7, DISPLAY	
43	22C0	R		MOV R1, R3	
44	8204	I		ADDI R1, 0x04	
45	73C0	R		ST R1, [R7]	DISPLAY = tag|4
46	1EF2	I		LDI R7, TEMP	
47	63C0	R		LD R1, [R7]	
48	1E05	I		LDI R7, V_TEMP	
49	73C0	R		ST R1, [R7]	log temperature (sensor read-back only)
4A	1E01	I		LDI R7, V_WASH	
4B	6BC0	R		LD R5, [R7]	
4C	1EF9	I		LDI R7, T_LIMIT	
4D	7BC0	R		ST R5, [R7]	LIMIT = wash ticks
4E	1EFA	I		LDI R7, T_CTRL	
4F	1203	I		LDI R1, 0x03	
50	73C0	R		ST R1, [R7]	clear COUNT/MATCH and enable
51	10F0	I		LDI R0, IN_STATUS	
52	1410	I		LDI R2, 0x10	COUNT bit4 -> MOTOR_DIR (toggles every 16 ticks)
53	1624	I		LDI R3, A_WASH	
54	1810	I		LDI R4, 0x10	TRIP bit mask
55	1AF8	I		LDI R5, T_COUNT	
56	1EFB	I		LDI R7, T_STATUS	
57	6340	R	wash	LD R1, [R5]	
58	5280	R		AND R1, R2	
59	32C0	R		ADD R1, R3	LOCK | MOTOR_EN | DIR
5A	7380	R		ST R1, [R6]	
5B	6200	R		LD R1, [R0]	
5C	5300	R		AND R1, R4	
5D	A2A6	I		BNEZ R1, trip_fault	hardware interlock tripped
5E	63C0	R		LD R1, [R7]	
5F	9257	I		BEQZ R1, wash	until MATCH
60	1220	I		LDI R1, A_LOCK	
61	7380	R		ST R1, [R6]	stop motor
62	1228	I		LDI R1, A_DRAIN	
63	7380	R		ST R1, [R6]	
64	1E04	I		LDI R7, V_TAG	
65	63C0	R		LD R1, [R7]	
66	8205	I		ADDI R1, 0x05	
67	1EF5	I		LDI R7, DISPLAY	
68	73C0	R		ST R1, [R7]	DISPLAY = tag|5
69	1E03	I		LDI R7, V_DRAIN	
6A	69C0	R		LD R4, [R7]	R4 = threshold
6B	1EF1	I		LDI R7, WATER_LEVEL	

Addr	Hex	Fmt	Label	Assembly	Comment
6C	63C0	R	drain	LD R1, [R7]	
6D	2500	R		MOV R2, R4	
6E	C440	R		SGEU R2, R1	R2 = (threshold >= WATER_LEVEL)
6F	946C	I		BEQZ R2, drain	
70	1E04	I		LDI R7, V_TAG	
71	63C0	R		LD R1, [R7]	
72	8206	I		ADDI R1, 0x06	
73	1EF5	I		LDI R7, DISPLAY	
74	73C0	R		ST R1, [R7]	DISPLAY = tag|6
75	1E02	I		LDI R7, V_SPIN	
76	6BC0	R		LD R5, [R7]	
77	1EF9	I		LDI R7, T_LIMIT	
78	7BC0	R		ST R5, [R7]	LIMIT = spin ticks
79	1EFA	I		LDI R7, T_CTRL	
7A	1203	I		LDI R1, 0x03	
7B	73C0	R		ST R1, [R7]	clear + re-arm
7C	122C	I		LDI R1, A_SPIN	
7D	7380	R		ST R1, [R6]	
7E	10F0	I		LDI R0, IN_STATUS	
7F	1810	I		LDI R4, 0x10	
80	1EFB	I		LDI R7, T_STATUS	
81	6200	R	spin	LD R1, [R0]	
82	5300	R		AND R1, R4	
83	A2A6	I		BNEZ R1, trip_fault	
84	63C0	R		LD R1, [R7]	
85	9281	I		BEQZ R1, spin	
86	1220	I		LDI R1, A_LOCK	
87	7380	R		ST R1, [R6]	motor + pump off, door still locked
88	1EFA	I		LDI R7, T_CTRL	
89	1202	I		LDI R1, 0x02	
8A	73C0	R		ST R1, [R7]	stop + clear timer
8B	1E04	I		LDI R7, V_TAG	
8C	63C0	R		LD R1, [R7]	
8D	8207	I		ADDI R1, 0x07	
8E	1EF5	I		LDI R7, DISPLAY	
8F	73C0	R		ST R1, [R7]	DISPLAY = tag|7
90	1EF3	I		LDI R7, DRUM_RPM	
91	63C0	R	wait_drum	LD R1, [R7]	
92	A291	I		BNEZ R1, wait_drum	unlock only after the drum has stopped
93	1240	I		LDI R1, A_DONE	
94	7380	R		ST R1, [R6]	BUZZER on, DOOR_LOCK off
95	1E04	I		LDI R7, V_TAG	
96	63C0	R		LD R1, [R7]	
97	8208	I		ADDI R1, 0x08	
98	1EF5	I		LDI R7, DISPLAY	
99	73C0	R		ST R1, [R7]	DISPLAY = tag|8 (done)
9A	1EF0	I	release	LDI R7, IN_STATUS	
9B	63C0	R	wait_rel	LD R1, [R7]	
9C	1401	I		LDI R2, 0x01	
9D	5440	R		AND R2, R1	
9E	A49B	I		BNEZ R2, wait_rel	wait until START is released
9F	B009	I		JMP idle_wait	buzzer stays on until the next START
A0	1200	I	door_abort	LDI R1, A_OFF	
A1	7380	R		ST R1, [R6]	unlock, everything off
A2	1EF5	I		LDI R7, DISPLAY	
A3	12E1	I		LDI R1, 0xE1	
A4	73C0	R		ST R1, [R7]	
A5	B09A	I		JMP release	
A6	1200	I	trip_fault	LDI R1, A_OFF	
A7	7380	R		ST R1, [R6]	software clears MOTOR_EN request (needed for re-arm)
A8	1EFA	I		LDI R7, T_CTRL	
A9	1202	I		LDI R1, 0x02	
AA	73C0	R		ST R1, [R7]	stop timer
AB	1EF5	I		LDI R7, DISPLAY	
AC	12EE	I		LDI R1, 0xEE	
AD	73C0	R		ST R1, [R7]	
AE	1EF0	I		LDI R7, IN_STATUS	
AF	1810	I		LDI R4, 0x10	
B0	63C0	R	wait_rearm	LD R1, [R7]	
B1	5300	R		AND R1, R4	
B2	A2B0	I		BNEZ R1, wait_rearm	TRIP clears only after door closed + new START edge

Addr	Hex	Fmt	Label	Assembly	Comment
B3	B009	I		JMP idle_wait	re-armed: START is held, a fresh cycle begins